

TECNOLOGIA SETÚBAL

ESCOLA SUPERIOR POLITÉCNICO SETÚBAL

AUTHOR'S FULL NAME

(Uppercase, Arial font, regular, 16 pt.)

PROJECT REPORT TITLE

(All uppercase, Arial, bold, 16 pt.)

Introduction to the Master's Project in Software Engineering Report

(Arial font, regular, 12 pt)

SUPERVISOR(S)

(Degree, Name)

(Arial font, regular, 12 pt)

Date (month and year)
(Arial font, regular, 10 pt)

Resumo

Insert in this section the text of the abstract in English. If an automatic translation tool is used, ensure that the translation is correct in this language; if necessary, ask someone to review the text.

Palavras-chave: Present a maximum of six keywords in Portuguese, separated by commas.

Abstract

The abstract should have a maximum of 100 words and must not exceed one page. It should concisely present the topic of the Master's Project, the problem to be addressed, and its relevance within the context of Software Engineering. It should also identify the general and specific objectives of the work, as well as the scientific and technical background that supports the proposed project.

The abstract should include a brief description of the proposed methodological approach, indicating the methods, techniques, or strategies that are expected to be used to achieve the defined objectives.

This report corresponds to the initial phase of the Master's Project, focusing on the review of the state of the art, the definition of the problem, and the planning of the work to be carried out. Final conclusions or experimental results should not be presented.

The use of abbreviations and acronyms should be avoided in the abstract. When considered indispensable, they must be clearly defined.

Keywords: Present a maximum of six keywords, separated by commas

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List of Acronyms and Abbreviations

ISDN	Integrated Services Digital Network
WiFi	Wireless Fidelity

(Sort alphabetically)

1 Introduction

The introduction should present the general framework of the Master's Project, contextualizing the topic within the field of Software Engineering and highlighting its scientific and/or technological relevance.

This chapter should provide the reader with an overall view of the problem to be addressed, the underlying motivations of the work, and the objectives to be achieved, as well as a brief description of the organization of the document.

The introduction should not anticipate results or final conclusions, as the project is still at an initial stage.

As an indicative guideline, this chapter should have an extent appropriate to the scope of the work, typically between two and three pages.

1.1 Context and Background

This section should present the academic, scientific, and/or technological context in which the project is developed.

The objective is to frame the topic within the field of Software Engineering, describing the general scenario in which the problem arises and its importance in the current context.

Among other aspects, the following may be addressed:

- the application domain of the project.
- relevant technological or scientific trends.
- challenges currently recognized in the area.
- organizational or industrial context, if the project is associated with a real-world scenario.

This section should help the reader understand where the problem is situated and why it is relevant to study it.

1.2 Problem Statement

This section should clearly identify and describe the problem that the project intends to address.

The problem statement should be concrete, well-defined, and understandable, avoiding overly vague or generic descriptions.

The problem should stem from the context presented previously and, whenever possible, be related to limitations, gaps, or challenges identified in practice or in the existing literature.

At this stage, a detailed solution is not expected, but rather a clear definition of the problem that justifies the development of the project.

1.3 Motivation and Relevance

The motivation should explain why the identified problem is worth investigating.

This section should highlight the factors that make the project relevant from a scientific, technical, social, or industrial perspective.

For example, the following aspects may be addressed:

- limitations of existing approaches.
- unmet needs.
- opportunities for improvement.
- potential impact of the expected results.

The relevance of the project should be justified in a well-founded manner, avoiding purely descriptive or subjective arguments.

This section should also briefly describe the different parts that make up the report and how they are articulated with each other.

1.4 Project Objectives

This section should present the objectives of the Master's Project.

It is recommended to distinguish between:

- the general objective, which reflects the main purpose of the work.
- the specific objectives, which detail the main steps or goals to be achieved.

The objectives should be clear, realistic, and consistent with the formulated problem.

At this stage, the objectives refer to the work to be carried out, not to results already

obtained.

When appropriate, the objectives may be presented in the form of a numbered list

1.5 Document Structure

This section should provide a brief description of the structure of the report.

The main chapters that compose the document should be identified, succinctly explaining the content of each one and how they relate to each other.

The objective is to guide the reader regarding the organization of the document and to facilitate its overall understanding.

2 State of the Art

The State-of-the-Art chapter aims to situate the project within the context of existing knowledge, presenting the main concepts, approaches, and relevant works related to the topic under study.

This chapter should demonstrate that the identified problem does not arise in isolation but rather results from the analysis of the existing scientific and technical literature.

At this stage of the project, an exhaustive literature review is not expected. Instead, a careful and well-founded analysis of the most relevant works should be carried out, sufficient to support the formulation of the problem and the proposed approach.

2.1 Fundamental Concepts and Theoretical Framework

This section should present the main concepts, definitions, and theoretical foundations necessary to understand the topic of the project.

The objective is to establish a common conceptual basis that allows the reader to understand the domain in which the work is situated.

Only the concepts that are relevant to the problem under study should be included, avoiding overly generic or extensive descriptions. Whenever appropriate, the concepts should be supported by suitable bibliographic references.

2.2 Related Work

This section should analyse scientific, technical, or applied works related to the topic of the project.

The analysis should focus on existing approaches, methods, solutions, or results that are relevant to the identified problem.

A simple enumeration of works is not intended; instead, a critical analysis should be provided, highlighting similarities, differences, advantages, and limitations of existing approaches, whenever relevant.

The selected works should be current and relevant, and may include scientific articles, dissertations, technical reports, or recognised standards in the field of Software Engineering.

2.3 Identified Gaps and Opportunities

Based on the analysis presented in the previous sections, this section should identify gaps, limitations, or opportunities for improvement in the existing literature.

The objective is to highlight how the proposed project differentiates itself from or complements existing works.

The identified gaps should be directly related to the problem formulated in Chapter 1 and should contribute to justifying the relevance and pertinence of the Master's Project.

At this stage, it is not necessary to present detailed solutions, but rather to clarify the space in which the project is positioned and the expected contribution of the work to be developed.

3 Proposed Approach

This chapter aims to present the proposed approach for the development of the Master's Project.

The methodological options considered, the methods and techniques expected to be used, as well as the scope and boundaries of the work, should be described in a clear and structured manner.

At this stage of the project, a detailed description of the implementation or the presentation of results is not expected. The focus should be on the coherence between the formulated problem, the defined objectives, and the proposed approach to achieve them.

3.1 Overview of the Approach

his section should present an overall view of the approach adopted in the project.

The objective is to provide a high-level description of the type of work to be carried out, allowing the reader to understand the general logic of the project.

Aspects that may be addressed include:

- the nature of the project (for example, applied research, prototype development, empirical study, case study, among others).
- the general strategy to be adopted to address the identified problem.
- the articulation between theoretical and practical components, when applicable.

This description should be concise and sufficiently clear to contextualise the following sections.

3.2 Proposed Methods and Techniques

This section should describe the methods, techniques, and tools that are expected to be used throughout the development of the project.

The description should be provided at a level appropriate to the initial phase of the work, without the need to detail implementation procedures.

By way of example, the following may be included:

- research methods (for example, literature review, case study, experimentation, qualitative or quantitative analysis).

- software engineering techniques (for example, modelling, iterative development, testing, usability evaluation).
- tools or technologies that are anticipated to be used.

Whenever possible, the methodological choices should be briefly justified in relation to the objectives of the project.

3.3 Scope and Assumptions

This section should clarify the scope of the project and the main assumptions considered.

The objective is to clearly define what is included in the work and, equally importantly, what is outside its scope.

Among other aspects, the following may be addressed:

- limits of the application domain.
- technical, temporal, or organizational constraints.
- assumptions adopted for the execution of the project.

This clarification contributes to a better understanding of the scope of the work and helps to manage expectations regarding the expected results.

3.4 Initial Limitations and Risks (optional)

This section may be included to identify potential limitations and risks associated with the proposed approach.

At this stage, an exhaustive analysis is not intended, but rather an initial reflection on possible difficulties that may influence the development of the project.

Risks of a technical, methodological, or organisational nature may be considered, as well as preliminary strategies for their mitigation.

4 Project Planning

This chapter aims to present the planning of the work to be carried out within the scope of the Master's Project.

The main phases of the project, the proposed schedule, the main milestones and deliverables, as well as any risks identified at this initial stage, should be described.

The presented planning should be realistic and consistent with the defined objectives, taking into account the duration of the project and the available resources. At this stage, the planning is necessarily preliminary and may be adjusted throughout the development of the work.

4.1 Project Phases

This section should identify and describe the main phases of the Master's Project.

Each phase should correspond to a coherent set of activities, contributing clearly to the achievement of the project objectives.

The description of the phases should be concise, indicating the purpose of each phase and its relationship with the overall development of the work. It is not necessary to detail tasks at an operational level.

Examples of possible phases include:

- further development of the state of the art.
- definition and consolidation of the proposed approach.
- development and implementation of the solution.
- evaluation and validation of the results.
- writing of the final report.

4.2 Schedule and Main Milestones

This section should present the proposed schedule for the execution of the project.

The schedule should indicate the temporal distribution of the phases identified previously, as well as the main project milestones.

The schedule may be represented in textual form or by using appropriate diagrams (for example, a Gantt chart), whenever this contributes to the clarity of the information presented.

The objective of this section is to demonstrate that the project is feasible within the available time frame and that there is a consistent plan for the work to be carried out.

4.3 Expected Deliverables

This section should identify the main deliverables expected throughout the project.

Deliverables may include, among others:

- intermediate documents.
- software artefacts.
- models, prototypes, or tools.
- the final Master's report.

For each deliverable, its purpose and the project phase in which it is expected to be produced should be succinctly indicated.

4.4 Risks and Mitigation Strategies (optional)

This section may be used to identify potential risks associated with the development of the project.

Risks may be of a technical, methodological, temporal, or organisational nature.

For each identified risk, a preliminary mitigation strategy should be presented whenever possible.

At this stage, an exhaustive analysis is not expected, but rather an initial reflection that demonstrates awareness of the challenges that may arise throughout the project.

5 Conclusions and Next Steps

This chapter aims to synthesise the work developed in the Introduction to the Master's Project report and to present the next steps planned for the continuation of the project.

Final conclusions or empirical results should not be presented, as the project is at an initial stage.

The conclusions should reflect on the context established, the formulated problem, the defined objectives, and the proposed approach, reinforcing the overall coherence of the work.

5.1 Summary of the Work Developed

This section should present a synthesis of the main aspects addressed throughout the report.

The following elements should be succinctly highlighted:

- the identified context and problem.
- the relevance and motivation of the project.
- the defined objectives.
- the proposed approach for the development of the work.
- the established planning.

This synthesis should allow the reader to obtain an overall view of the project without the need to consult the previous chapters.

5.2 Next Steps

This section should describe the next steps planned for the development of the Master's Project.

The activities that will follow the planning phase presented in this report should be identified, establishing a clear link with the work to be developed in the final Master's report.

For example, the following may be mentioned:

- further development or consolidation of the state of the art.
- development and implementation of the proposed solution.
- definition of evaluation and validation strategies.
- writing and consolidation of the final report.

The objective of this section is to demonstrate continuity and clarity regarding the future evolution of the project.

References

In the preparation of the bibliographic references, one of the following styles may be used:

- the **author–date style** (for example, APA guidelines), or
- the **numerical style**, in accordance with the **IEEE** standard.

The selected style must be used consistently throughout the entire document, both in the in-text citations and in the list of references.

To facilitate the management of bibliographic references, the use of appropriate software is recommended, such as **Zotero** (https://www.zotero.org).

Only the sources that are effectively cited in the text of the report should be included in this section. All in-text citations must have a corresponding entry in this section, and vice versa.

Corporation for Digital Scholarship. (2023). Zotero (Version 6.x) [Computer software]. https://www.zotero.org/

Appendix A – Formatting Guidelines and Examples

(This chapter is purely illustrative and is intended to exemplify the use of sections, subsections, figures, and tables. Its content is not part of the scientific report and must be removed from the final version.)

1 Chapter 1

Illustrative introductory text exemplifying the beginning of a chapter.

1.1 Section 1

Illustrative example text.

All figures and tables must be referenced in the text of the report, as this reference is essential to understand how they fit into what is being described. Example: a Figure 1.1 illustrates technologies present in an urban environment.

1.2 Section 1.2

Illustrative text.

1.2.1 Subsection 1.2.1

Illustrative text for a subsection.

1.2.1.1 SUBSUBSETION 1.2.2-1

Example of an in-text citation using the author–date style (APA):

The use of bibliographic reference managers contributes to the consistency and correctness of references (Corporation for Digital Scholarship, 2023).



Figure 1.1 – Caption of the First Figure of Chapter 1.

Note: When figures from other authors are used, the source must be referenced in the caption – see Figure 1.2.



Figure 1.2 - Smart Cities (adapted from Norman, 2013).

1.3 Section 3

Example of an in-text reference: Table 1.1 presents an example of a table.

Table 1.1 − First Table of Chapter 1.

A	**B**	**C**
A1	B1	C1
A2	B2	C2